

AgriShare

Technical System Overview & Architecture

A comprehensive engineering specification of the AgriShare peer-to-peer agricultural equipment sharing marketplace. Detailing the Next.js frontend, Django REST Framework backend, relational database schema, JWT auth flow, and automated testing architecture.

Next.js 16 App Router

React 19 & TypeScript

Django 5.1 & DRF

SimpleJWT Token Auth

SQLite / PostgreSQL

OpenAPI 3 / Swagger

Vanilla CSS Design System

MONOREPO SPECIFICATION

Engineering Specification Rev 2.0

TARGET DEPLOYMENT

Containerized Full-Stack Cloud

Executive Summary & Problem Statement

Addressing the structural capital expenditure deficit across Indian agricultural communities.

The Agricultural Dilemma

Over 85% of Indian farmers cultivate operational holdings under 2 hectares. High capital machinery like 50+ HP 4WD tractors (₹8L–₹12L) and combine harvesters (₹25L–₹35L) are financially inaccessible to buy outright.

Meanwhile, machinery owners in progressive belts suffer from **65% seasonal idle time**, where machines sit depreciating in sheds between sowing and harvest cycles.

The AgriShare Solution

AgriShare establishes an automated, trust-backed peer-to-peer equipment sharing marketplace. By disintermediating local informal brokers, AgriShare democratizes mechanization:

- Zero middleman brokerage markup
- Verified machinery condition and transparent security deposits
- Role-based dashboards for renters and machine fleet owners

Core Engineering Design Principles

Architectural pillars guiding the entire codebase and infrastructure development.



High Efficiency & Low Latency

Built without bloated CSS frameworks; pure Vanilla CSS with zero runtime JavaScript styling overhead ensures fast loading even over rural 3G/4G connections.



Domain Isolation & Modularity

Backend segmented into 6 micro-apps (accounts, equipment, bookings, reviews, notifications, dashboard) with loose coupling.



Strict Transactional Safety

Database-level exclusion constraints prevent overlapping bookings, while state machines enforce rental transitions and reviews.

Monorepo Layout Summary

```
AgriShare/ |— backend/ (Django 5.1 REST API + Custom Auth + Seeders)
           |— frontend/ (Next.js 16 App Router + TypeScript + Design Tokens)
           └— scripts/ (Automated E2E Flow Tests & Shell Launchers)
```

Comprehensive Full-Stack Technology Matrix

Detailed inventory of software libraries, frameworks, and system responsibilities.

Layer	Technology & Version	Purpose & Responsibility	Key Architectural Benefit
Frontend	Next.js 16+	Application Router, Server & Client Components	Optimized SSR/CSR rendering, fast navigation
Frontend	TypeScript 5.x	Strict end-to-end static type safety	Eliminates runtime undefined & API schema drifts
Frontend	Vanilla CSS Tokens	Luxury Forest & Harvest Amber Design System	0KB runtime CSS overhead, maximum layout control
Frontend	Lucide React	Clean vector iconography	Consistent visual language, tree-shakeable icons
Backend	Python 3.12+ / Django 5.1	Core Web Framework & ORM Engine	Robust migrations, admin portal, rapid prototyping
Backend	Django REST Framework	RESTful API endpoints, serializers, viewsets	Automated serialization, pagination, content negotiation
Backend	djangorestframework-simplejwt	JWT Token authentication & sliding refresh	Stateless auth, secure Bearer token rotation
Backend	drf-spectacular	OpenAPI 3.0 generation & Swagger UI	Interactive API documentation at /api/docs/
Database	SQLite (Dev) / PostgreSQL (Prod)	Relational ACID persistence layer	Zero setup locally; zero schema changes for production

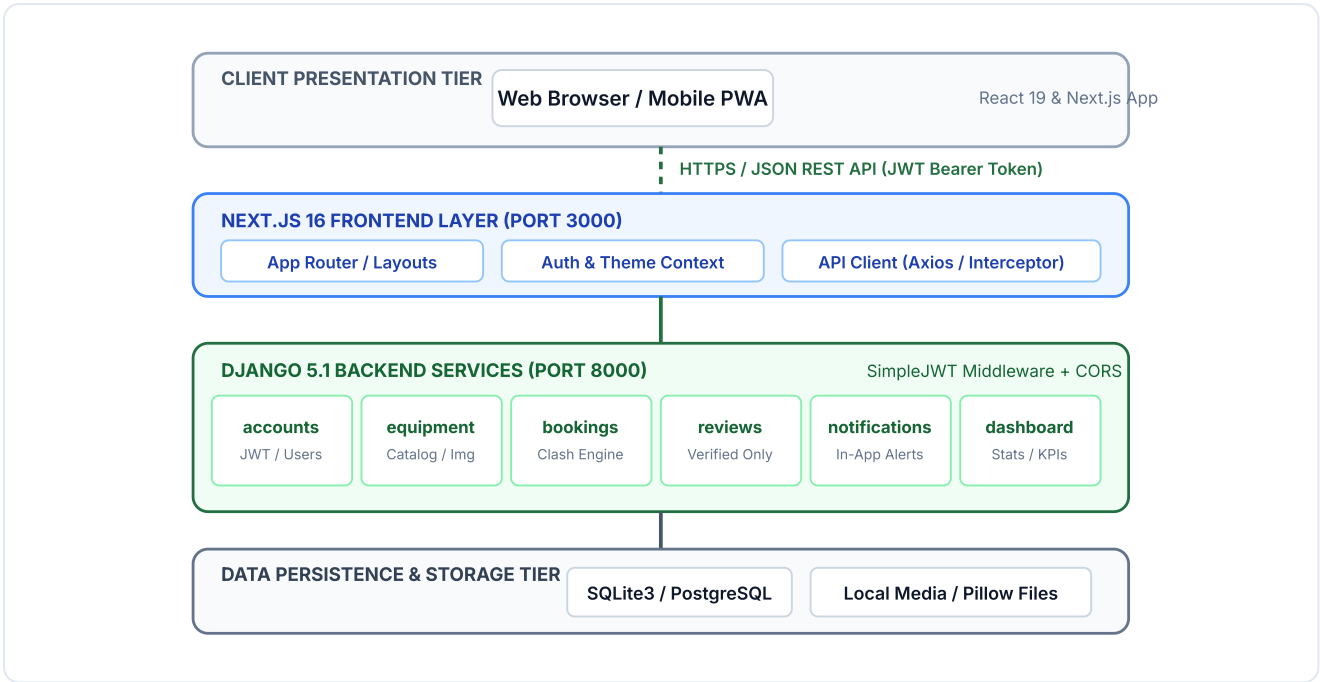
AgriShare Design System & Token Catalog

Tailored Indian agricultural palette designed for high contrast and visual elegance.

Forest Green #236f41 Primary CTA, Badges	Harvest Amber #d97706 Deposits, Warnings	Deep Slate #0f172a Headings & Dark Mode	Surface Slate #f8fafc Card & Backgrounds
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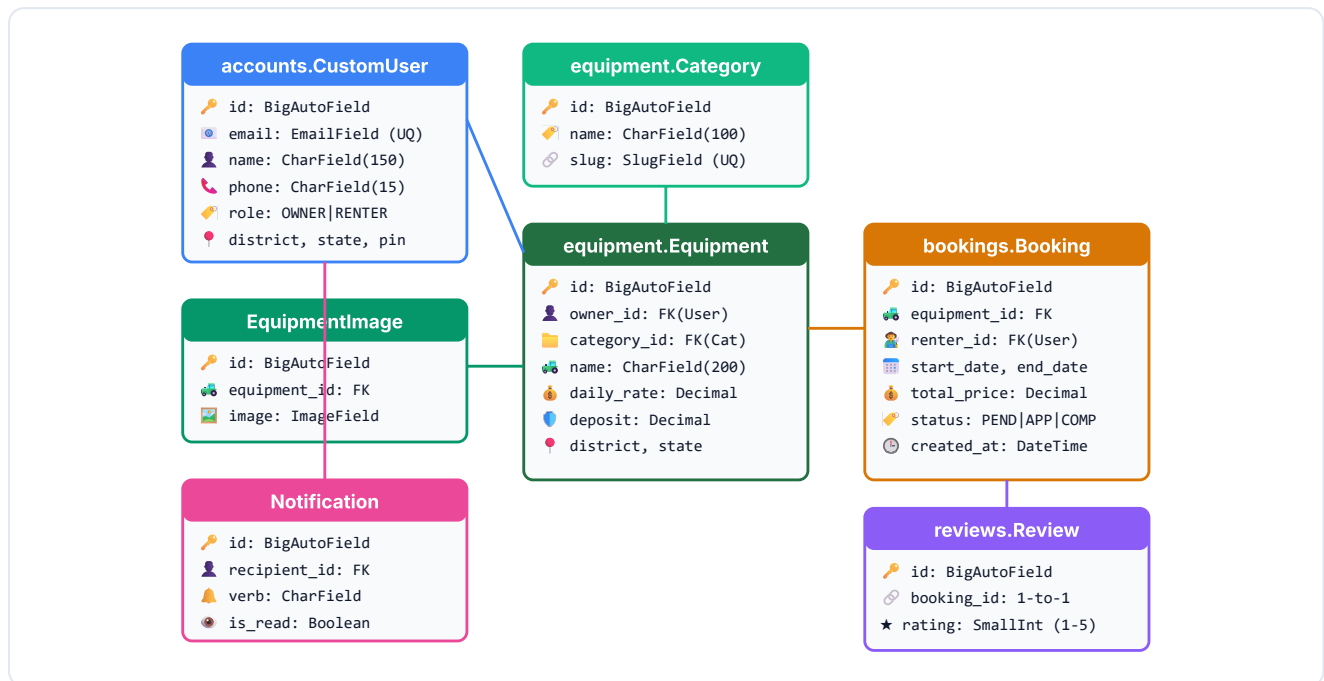
Layered System Architecture

Multi-tier client-server topology illustrating data flow, security boundaries, and micro-app decomposition.



Database Schema & Entity-Relationship Model

Normalized relational data structures enforcing business rules, constraints, and audit trails.



Authentication & Security Architecture

Stateless JWT session management, automatic token refresh, and authorization checks.



Primary REST API Endpoint Directory

Exhaustive matrix of core API routes exposed by the Django REST Framework backend.

Method	Endpoint URI	App Scope	Auth Policy	Description
POST	/api/auth/register/	accounts	Public	Creates user with role (OWNER/RENTER)
POST	/api/auth/login/	accounts	Public	Validates credentials & issues JWT pair
GET	/api/equipment/	equipment	Public	Paginated catalog with multi-filters
POST	/api/equipment/	equipment	Owner Only	Registers machinery with multi-photos
POST	/api/bookings/	bookings	Renter Only	Initiates booking; checks date overlaps
POST	/api/bookings/{id}/approve/	bookings	Owner Only	Approves request; reserves calendar
POST	/api/bookings/{id}/complete/	bookings	Owner/Admin	Marks returned; triggers review eligibility
POST	/api/reviews/	reviews	Verified Renter	Submits rating; updates machine average
GET	/api/dashboard/stats/	dashboard	Authenticated	Aggregates earnings or expense metrics

Automated Testing & Quality Metrics

Rigorous test coverage validating authorization barriers, date math, and state integrity.

19 Tests

Django Unit Suite

100% passing across accounts, equipment, and bookings.

E2E Flow

Automated Python Test

Simulates signup → search → book → approve → review.

0 Errors

Next.js Production Build

TypeScript compile passed with zero hydration warnings.

Engineering Roadmap & Future Horizons

Strategic evolution from functional MVP to nationwide connected agricultural infrastructure.

Phase 1: UPI & Escrow Smart Contracts (Q1 2027)

IN PLANNING

Direct UPI / Razorpay integration with automated escrow deposit locks that release upon digital signature verification.

Phase 2: IoT Telematics & GPS Geo-Fencing (Q2 2027)

ARCHITECTURE READY

OBD-II hardware sensor tracking for live tractor engine run-hours, fuel consumption analytics, and virtual field boundary geo-fencing.

Phase 3: AI Crop-Harvest Machinery Matching (Q3 2027)

RESEARCH

Predictive machine learning models forecasting regional harvester demand based on satellite weather data and crop maturity cycles.

Phase 4: Multilingual Voice Assistant (Q4 2027)

R&D

Voice-driven booking interface supporting Hindi, Punjabi, Gujarati, and Marathi for illiterate and semi-literate rural operators.